

國立成功大學
電機工程學系
碩士論文

基於因果感知機制之時空預訓練語言模型
於缺失值補全

**CauST-PLM: A Causality-aware Spatio-Temporal
Pre-trained Language Model for Imputation**

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中華民國一百一十五年七月

國立成功大學

National Cheng Kung University

碩士論文合格證明書

Certificate of Approval for Master's Thesis

題目：基於因果感知機制之時空預訓練語言模型於缺失值補全

Title: CauST-PLM: A Causality-aware Spatio-Temporal Pre-trained
Language Model for Imputation

研究生：蔡佳蓁

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基於因果感知機制之時空預訓練語言模型於缺失值補全

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摘要

時空資料補值對於資料驅動應用的可靠性至關重要，因為現實世界中由分散式感測器所蒐集的觀測資料，經常因設備故障、通訊錯誤或異常資料傳輸而產生缺失。現有方法雖然能在一定程度上建模時間模式或空間相關性，但在高缺失率、長距離時空依賴以及感測器間雜訊相關性的情境下，仍面臨顯著挑戰。近年來，預訓練語言模型在序列建模上展現出優異能力，然而由於時空資料具有數值化、圖結構化以及對缺失模式敏感等特性，使得直接將 PLMs 應用於時空補值任務仍存在困難。本研究提出 CauST-PLM，一種用於資料補值之因果感知時空預訓練語言模型。CauST-PLM 首先將不完整的時空觀測轉換為具遮罩感知的時間與空間表徵單元，使 PLM 能夠明確區分已觀測值與缺失值。接著，引入硬提示作為文字前綴，以協助 PLM 對齊補值目標與缺失值表示規則。為了提升空間表示品質，本研究設計了一個輕量化的時空因果注意力模組，在 PLM 推理前先過濾雜訊或弱相關的感測器資訊，以提升空間資訊表徵能力。在四個公開時空交通資料集（PEMS03、PEMS04、PEMS07 與 PEMS08）上的大量實驗結果顯示，CauST-PLM 在不同隨機缺失率設定下皆能展現良好的補值表現。與多種代表性基準方法相比，CauST-PLM 在大多數實驗設定中取得較低的誤差，而消融實驗亦驗證了因果空間過濾機制與提示引導策略的有效性。

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關鍵字：時空資料、資料補值、預訓練語言模型、因果注意力



CauST-PLM: A Causality-aware Spatio-Temporal Pre-trained Language Model for Imputation

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Abstract

Spatio-temporal data imputation is essential for reliable data-driven applications, since real-world observations collected from distributed sensors are often incomplete due to device failures, communication errors, and abnormal data transmission. Existing methods can model temporal patterns or spatial correlations to some extent, but they still struggle with high missing rates, long-range spatio-temporal dependencies, and noisy correlations among sensors. Recently, pre-trained language models (PLMs) have shown strong sequence modeling ability, yet directly applying them to spatio-temporal imputation remains challenging because such data is numerical, graph-structured, and sensitive to missing-value patterns.

In this paper, we propose CauST-PLM, a causality-aware spatio-temporal pre-trained language model for imputation. CauST-PLM first transforms incomplete spatio-temporal observations into mask-aware temporal and spatial tokens, enabling the PLM to distinguish observed values from missing entries. A textual prefix prompt is further introduced to align the PLM with the imputation objective and the missing-value convention. To improve spatial representation quality, we design a lightweight spatial-temporal causal attention module that filters noisy or weakly relevant sensor information before PLM reasoning. Extensive experiments on four public spatio-temporal datasets, PEMS03, PEMS04, PEMS07, and PEMS08, demonstrate that CauST-PLM achieves strong imputation performance under different Random Point Missing rates. Compared with representative baselines, CauST-PLM obtains lower

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errors in most settings, and ablation studies verify the effectiveness of both causal spatial filtering and prompt guidance.

Keywords : Spatio-Temporal Data, Data Imputation, Pre-trained Language Models, Causal Attention



誌謝

時光飛逝，轉眼間兩年的碩士生活已悄然來到尾聲。這一路走來有挑戰、有挫折，也有許多值得珍惜的回憶。能夠順利完成這篇論文，並非僅靠一己之力，而是因為一路上有許多人的鼓勵、陪伴與協助，在此謹向所有曾經幫助過我的人，致上最誠摯的感謝。

首先，我要衷心感謝我的指導教授解巽評教授。在研究期間，老師始終耐心的指導，引導我一步步學習如何思考問題、分析問題並解決問題。老師給予我許多寶貴的啟發，讓我獲益良多。同時，也要誠摯感謝各位口試委員撥冗審閱本論文，並於口試中提供許多寶貴的建議與指教，使本論文得以更加完善，在此致上由衷的感謝。

接著，我想感謝一路陪伴我的家人。謝謝你們給予我最大的支持與鼓勵，成為我最堅強的後盾。同時，也要感謝實驗室博班學姐、同學以及朋友們，在研究與生活中給予我的幫助與陪伴。一起討論研究、互相解決程式問題，在忙碌之餘聊天、吃飯、分享生活點滴，讓原本充滿壓力的研究生活增添了許多溫暖與歡笑。

另外，我也想感謝自強校區夜晚遇見的那些可愛狗狗們。每當心情感到疲憊時，到校園散步、看看牠們、摸摸牠們，總能讓緊繃的心情獲得短暫的放鬆，也重新找回繼續努力的動力。對我而言，牠們也是支持我一路走到今天的重要力量之一。

最後，我想感謝這兩年努力沒有放棄的自己。雖然研究的過程並非一帆風順，曾經懷疑自己、迷惘過，也曾因一次次失敗而感到挫折，但因為經歷了這些過程，我才得以不斷成長，學會面對困難、培養獨立思考與解決問題的能力。願自己在未來的道路上，仍能保持對知識的熱情、對生活的好奇，以及面對未知挑戰的勇氣。

研究生 蔡佳蓁

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Chapter1 Introduction

Spatio-temporal data plays a fundamental role in many real-world applications, such as transportation systems, environmental monitoring, energy management, and urban computing. In distributed sensor networks, however, measurements are often incomplete due to sensor failures, communication errors, maintenance, and abnormal data transmission. These missing values can distort the observed dynamics and further degrade downstream tasks that rely on complete and reliable spatio-temporal signals. Therefore, accurate spatio-temporal data imputation is an essential preprocessing and modeling problem for robust data-driven analysis.

Existing imputation methods have made substantial progress, but they still face important limitations. Recurrent models such as BRITS [1] can capture sequential dependencies, yet they often struggle with long-range temporal patterns and complex spatial interactions. Generative methods, including CSDI [16] model uncertainty in missing values, but they usually require higher computational cost. Transformer-based imputation methods such as SAITS [6] improve long-range dependency modeling through self-attention mechanisms. However, most of them primarily focus on temporal representations and do not explicitly model graph-structured spatial dependencies among sensors. Recently, pre-trained language models (PLMs) have demonstrated remarkable sequence modeling and reasoning capabilities beyond natural language processing. Recent studies, such as Time-LLM [10] and LLM4TS[2], further show that PLMs can be effectively adapted to time-series analysis by bridging the modality gap between numerical observations and language through representation alignment techniques, including tokenization, prompt design, and feature reprogramming.

Nevertheless, directly applying PLMs to spatio-temporal imputation remains challenging because spatio-temporal observations are continuous, graph-structured numerical data with irregular missing patterns, which differ substantially from the textual data used during PLM pre-training. Therefore, effectively encoding spatial structure, temporal dynamics, and missing-value information into PLM-compatible representations remains a critical challenge.

The main challenges of applying PLMs to spatio-temporal imputation can be summarized as follows.

First, an **effective representation alignment mechanism** is required to transform incomplete graph-structured spatio-temporal observations into PLM-compatible token representations while preserving temporal dynamics, spatial dependencies, and missing-value information.

Second, **not all spatial correlations are equally informative**. Noisy or spurious sensor relationships may interfere with the representations used for imputation, especially under high missing rates.

Third, **PLMs require task-specific semantic guidance** to better align their reasoning process with the imputation objective and missing-value convention.

To address these challenges, we propose **CauST-PLM**, a causality-aware spatio-temporal pre-trained language model for imputation. To address the **first challenge**, we adopt a **mask-aware spatio-temporal tokenization** strategy to convert incomplete observations into spatial and temporal tokens, enabling the PLM to distinguish observed values from missing entries. To address the **second challenge**, we introduce a lightweight **Spatial-temporal Causal Attention (SCA)** module to suppress noisy or weakly relevant sensor information before PLM reasoning, allowing the model to focus on more informative spatial dependencies. To address the **third challenge**, we prepend a **Textual Prefix Prompt** that provides task-level

guidance about the imputation objective and the missing-value convention, helping the PLM better align its reasoning process with the imputation task. By integrating these components, CauST-PLM improves the robustness of PLM-based spatio-temporal imputation under different missing rates.

The main contributions of this work are summarized as follows:

- We propose CauST-PLM, a PLM-based framework for spatio-temporal data imputation that integrates mask-aware spatio-temporal tokenization, textual prefix prompting, and output projection for reconstructing missing values.
- A causality-aware spatial filtering module based on Spatial Causal Attention (SCA), which leverages causal relevance scores to refine spatial tokens by suppressing spurious correlations and emphasizing informative spatial dependencies.
- We conduct extensive experiments on four public traffic flow datasets, PEMS03, PEMS04, PEMS07, and PEMS08, under multiple Random Point Missing rates. The results demonstrate that CauST-PLM achieves competitive performance and the best overall trade-off across different missing scenarios.

Chapter2 Related work

2-1. Spatio-Temporal Data Imputation

Spatio-temporal imputation has evolved from statistical methods to deep learning-based approaches. Early recurrent models such as BRITS [1] capture temporal dependencies but struggle with long-range patterns and complex spatial interactions. Transformer-based imputation methods such as SAITS [6] improve long-range dependency modeling through self-attention, but mainly focus on temporal representations and often underexploit graph-structured spatial dependencies. Graph-based approaches, notably GRIN [5] explicitly model sensor relationships for missing value recovery, improving spatial dependency modeling. More recently, diffusion-based methods such as CSDI [16] model uncertainty in missing values through iterative denoising, while PriSTI [13] further extends this idea to spatio-temporal settings by incorporating graph-aware conditional diffusion. Although these methods improve imputation quality, diffusion-based models usually incur higher computational cost due to multi-step sampling.

Recently, pre-trained language models (PLMs) have been explored for time-series modeling. Representative PLM-based approaches, including Time-LLM [10] and GPT4TS [18], adapt PLMs to numerical sequences and demonstrate strong sequence modeling capabilities. However, these approaches typically treat each sensor as an independent sequence and lack explicit modeling of spatial dependencies and graph structure, limiting their effectiveness in spatio-temporal settings.

2-2. Bottleneck Attention

Handling large numbers of spatial tokens remains a key challenge for PLM-based models. Recent works introduce bottleneck or token compression mechanisms to reduce computational complexity and capture higher-order dependencies. Similar latent bottleneck designs have been explored in Perceiver IO [9], where a fixed set of learnable latent queries attends to high-dimensional input tokens for efficient representation learning.

For spatio-temporal modeling, sandglass-style attention further aggregates node-level representations into a smaller set of latent tokens, enabling efficient modeling of global interactions while preserving essential information. Similar ideas have also been explored in prior works such as SSTBAN [7] and CrossFormer [17], which utilize learnable reference representations combined with attention mechanisms to model higher-order correlations while improving computational efficiency.

These approaches demonstrate that structured token compression is effective for scaling attention-based models to large spatio-temporal graphs.

2-3. Causality-Aware Modeling

Causality-aware methods aim to improve robustness by distinguishing meaningful dependencies from spurious correlations. In time-series analysis, causal discovery frameworks, exemplified by CUTS [4] have shown that identifying causal relationships can improve representation quality and robustness by filtering irrelevant or indirect dependencies.

Building upon this intuition, recent spatio-temporal models incorporate causal mechanisms into representation learning. For example, Casper introduces Spatial Causal Attention

(SCA), which aims to suppress spatial interactions that are likely induced by spurious correlations through causal-aware filtering. These studies suggest that filtering spatial representations before downstream reasoning can improve robustness under incomplete observations.

However, existing causality-aware methods are mainly designed for graph neural networks and have not been effectively integrated with pre-trained language models. This motivates us to investigate how causal-inspired spatial filtering can be incorporated into a PLM-based spatio-temporal imputation framework.



Chapter3 Preliminaries

3-1. Problem Definition

We consider a sensor network represented as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V}$ denotes a set of N sensors and $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the adjacency matrix encoding spatial dependencies among sensors.

Let $\mathbf{X} \in \mathbb{R}^{T \times N \times C}$ denote the complete spatio-temporal tensor, where T is the sequence length, N is the number of sensors, and C is the number of features.

A binary mask tensor $\mathbf{M} \in \{0, 1\}^{T \times N \times C}$ is used to indicate the observation status of each entry, where $m_{t,n,c} = 1$ denotes an observed value and $m_{t,n,c} = 0$ denotes a missing value. The mask tensor is provided together with the incomplete observations to explicitly indicate the locations of missing values, allowing the subsequent model to distinguish observed and missing entries during representation learning. The partially observed tensor is defined as

$$\tilde{\mathbf{X}} = \mathbf{X} \odot \mathbf{M}, \quad (3.1)$$

where $\odot$ denotes the element-wise product.

3-2. Task Objective

Given the partially observed tensor $\tilde{\mathbf{X}}$, the corresponding mask $\mathbf{M}$, and the graph structure $\mathcal{G}$, the goal of spatio-temporal imputation is to learn a function $\mathcal{F}$ that reconstructs the missing entries in $\mathbf{X}$:

$$\hat{\mathbf{X}} = \mathcal{F}(\tilde{\mathbf{X}}, \mathbf{M}, \mathcal{G}), \quad (3.2)$$

where $\hat{\mathbf{X}} \in \mathbb{R}^{T \times N \times C}$ is the imputed tensor.

During training, the model is optimized to minimize the reconstruction error on the missing entries. During evaluation, performance is measured only on entries where $m_{t,n,c} = 0$.



Chapter4 Methodology

We propose the Causality-aware Spatio-Temporal Pre-trained Language Model (CauST-PLM) for spatio-temporal imputation. As illustrated in Figure 4.1, the proposed framework consists of three major components.

(1) **Representation Construction.** Incomplete observations are transformed into mask-aware spatial and temporal tokens by combining temporal embeddings, node embeddings, and missing-value patterns, providing PLM-compatible representations.

(2) **Causality-aware PLM Reasoning.** The spatial tokens are first refined by the proposed Spatial-temporal Causal Attention (SCA) module to suppress noisy correlations. A textual prefix prompt and a bottleneck encoder-decoder are further incorporated to facilitate efficient PLM reasoning.

(3) **Reconstruction.** The refined representations are projected back to the node level and reconstructed into complete spatio-temporal observations through the output projection and reconstruction objective.

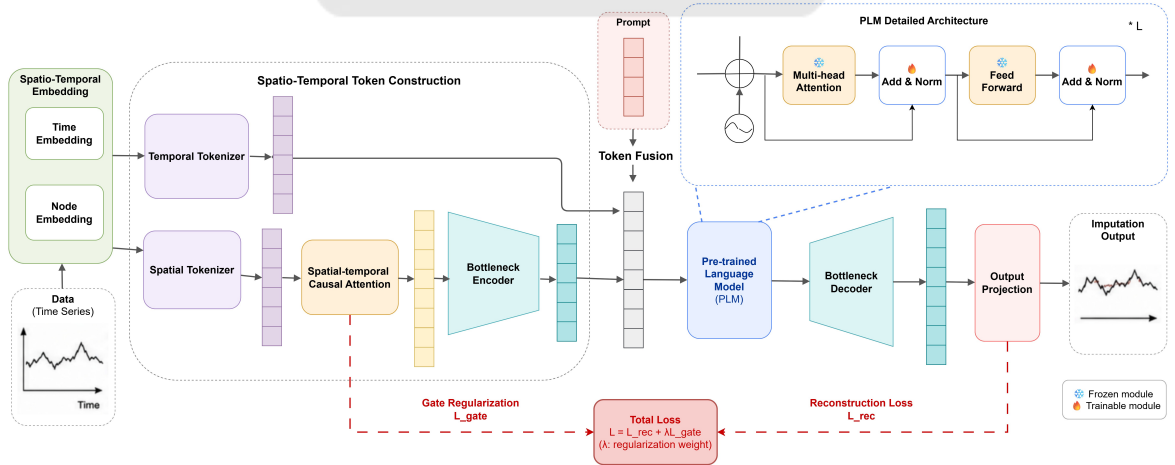


Figure 4.1: Overall architecture of the proposed model.

4-1. Time and Node Embedding

To enhance the representation of spatio-temporal data, we introduce both time embeddings and node embeddings to encode temporal periodicity and spatial structural information.

Time Embedding: Traffic data exhibit strong temporal periodicity, which is essential for imputation. Following common traffic patterns, we consider two periodic factors: time-of-day and day-of-week. Accordingly, we construct two learnable embedding tables, $E_{day} \in \mathbb{R}^{288 \times d_t}$ and $E_{week} \in \mathbb{R}^{7 \times d_t}$, where d_t denotes the temporal embedding dimension. For each timestamp, the time-of-day index is computed as:

$$s_t = \left\lfloor \frac{hour_t \times 60 + minute_t}{5} \right\rfloor \quad (4.1)$$

The final temporal embedding is obtained by concatenating the corresponding embeddings:

$$e_t = [E_{day}(s_t); E_{week}(w_t)] \quad (4.2)$$

For the input window, we use the temporal embedding of the last observed time step, denoted as e_T , as the global temporal context.

Node Embedding: To encode spatial identity and graph structural information, we construct node embeddings from the road network adjacency matrix.

Given the adjacency matrix A , we first compute the normalized graph Laplacian:

$$L = I - D^{-1/2} A D^{-1/2} \quad (4.3)$$

where D is the degree matrix.

We then perform eigen-decomposition on L and select the top- k eigenvectors as structural descriptors for each node. These descriptors are projected into the embedding space through a learnable linear transformation:

$$NE = \text{Linear}(U_k) \quad (4.4)$$

where $U_k \in \mathbb{R}^{N \times k}$ denotes the selected Laplacian eigenvectors, and $NE \in \mathbb{R}^{N \times d_n}$ is the resulting node embedding matrix. Specifically, $\mathbf{NE}_n$ denotes the node embedding of sensor n .

4-2. Spatio-Temporal Tokenization

To bridge spatio-temporal tensors and PLM-compatible embeddings, CauST-PLM first converts the partially observed history into two types of continuous tokens: node-level spatial tokens and global temporal tokens. Given an input window $\tilde{\mathbf{X}} \in \mathbb{R}^{T \times N \times C}$ and its observation mask $\mathbf{M} \in \{0, 1\}^{T \times N \times C}$, we concatenate the values and masks along the feature dimension so that each token is aware of both the observed state and its missing pattern:

$$\mathbf{U}_n = \text{Flatten}_{t,c}([\tilde{\mathbf{X}}_{:,n,:} \parallel \mathbf{M}_{:,n,:}]) \quad (4.5)$$

For spatial tokenization, each sensor n is represented as one PLM token by combining the masked observations, missing-value patterns, and the corresponding spatio-temporal embeddings. The spatial token is then computed as

$$\mathbf{s}_n = \text{LN}(\mathbf{W}_x \mathbf{U}_n + \mathbf{W}_m \text{Flatten}(\mathbf{M}_{:,n,:}) + \text{MLP}_s([\mathbf{e}_T \parallel \mathbf{NE}_n])) \quad (4.6)$$

where $\mathbf{W}_x$ projects the masked historical observations, $\mathbf{W}_m$ explicitly injects the missingness pattern, and $\text{LN}(\cdot)$ denotes layer normalization. This design produces $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_N] \in$

$\mathbb{R}^{N \times D}$, preserving one token per sensor before causal ordering and bottleneck compression.

For temporal tokenization, CauST-PLM summarizes the network-wide temporal dynamics rather than generating one token per timestamp. We first average the value-mask features over all sensors:

$$\bar{\mathbf{u}}_t = \frac{1}{N} \sum_{n=1}^N [\tilde{\mathbf{X}}_{t,n,:} \parallel \mathbf{M}_{t,n,:}]. \quad (4.7)$$

Based on the aggregated temporal representation, two complementary temporal tokens are generated: a state token that represents the overall system condition within the input window, and a gradient token that captures short-term temporal variations:

$$\mathbf{t}_{state} = \text{MLP}_{state}([\text{Flatten}(\bar{\mathbf{u}}_{1:T}) \parallel \mathbf{e}_T]), \quad (4.8)$$

$$\mathbf{t}_{grad} = \text{MLP}_{grad}([\text{Flatten}(\bar{\mathbf{u}}_{2:T} - \bar{\mathbf{u}}_{1:T-1}) \parallel \mathbf{e}_T]). \quad (4.9)$$

The state and gradient tokens are then concatenated and normalized to form the temporal token representation:

$$\mathbf{T}_{tok} = \text{LN}([\mathbf{t}_{state}; \mathbf{t}_{grad}]), \quad (4.10)$$

where $[\cdot; \cdot]$ denotes token-wise concatenation and $\text{LN}(\cdot)$ denotes layer normalization. The resulting $\mathbf{T}_{tok}$ is used as the global temporal context for subsequent PLM reasoning.

The generated temporal and spatial tokens serve as the initial representations for subsequent processing. The spatial tokens are first refined by the proposed Spatial-temporal Causal Attention (SCA) module and compressed by the bottleneck encoder, while the temporal tokens are retained as global temporal context. These representations are then combined with the textual prompt and fed into the PLM for reasoning.

4-3. Spatial-temporal Causal Attention (SCA)

Although attention mechanisms are effective for collecting contextual information, high attention scores only indicate statistical associations and do not necessarily imply causal relevance. In spatio-temporal imputation, corrupted or indirectly correlated sensors may introduce spurious messages into the representations used for reconstruction, especially under high missing rates.

Casper addresses this problem from a causal perspective. Motivated by structural causal modeling, it argues that conventional graph attention may capture spurious spatial correlations introduced by latent confounders rather than truly informative dependencies. Therefore, instead of directly relying on attention scores, Casper introduces a causality-aware spatial filtering mechanism whose objective is to suppress non-causal spatial interactions before downstream representation learning.

Motivated by this observation, we incorporate the idea of causal-inspired spatial filtering into our PLM-based framework. Instead of reproducing the complete graph-oriented causal modeling framework adopted in Casper, we preserve its core objective of suppressing spurious spatial correlations before representation learning and reformulate it as a lightweight token-level spatial filtering module implemented through residual gating.

Specifically, rather than learning a dense pairwise spatio-temporal attention map over all sensors and timestamps, our SCA estimates whether each node token should participate in the current reconstruction context. Given spatial tokens $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_N]$, we first summarize the global spatio-temporal field by mean pooling:

$$\mathbf{q} = \frac{1}{N} \sum_{n=1}^N \mathbf{s}_n. \quad (4.11)$$

The query $\mathbf{q}$ represents the current network-level state and serves as the target context for deciding which sensors are causally informative. For each sensor token, SCA estimates a query-aware gate by concatenating the local node representation with the global query:

$$\beta_n = \sigma(\text{MLP}_{sca}([\mathbf{s}_n \parallel \mathbf{q}])), \quad (4.12)$$

where $\beta_n \in (0, 1)$ is a soft causal relevance. A larger β_n means that the node token is more likely to provide useful causal relevance for the current global state, while a smaller value suppresses noisy or weakly relevant sensors. The gated spatial token is obtained by residual scaling:

$$\tilde{\mathbf{s}}_n = (\epsilon + (1 - \epsilon)\beta_n) \mathbf{s}_n, \quad (4.13)$$

$$\mathbf{S}' = \{\tilde{\mathbf{s}}_1, \tilde{\mathbf{s}}_2, \dots, \tilde{\mathbf{s}}_N\} \quad (4.14)$$

where ϵ prevents potentially useful node information from being completely removed. In our implementation, $\epsilon = 0.2$. This design can be viewed as a differentiable relaxation of a binary causal selector: rather than sampling a hard gate, the model learns a continuous causal relevance score and uses it to modulate the token magnitude. We further add a small sparsity regularization term $\lambda_g \frac{1}{N} \sum_{n=1}^N \beta_n$ to discourage uniformly high gates, encouraging the model to focus on a compact set of causally relevant sensors. The resulting gated spatial tokens $\tilde{\mathbf{S}} = [\tilde{\mathbf{s}}_1, \dots, \tilde{\mathbf{s}}_N]$ are then passed to the bottleneck attention module. The proposed SCA is causal-inspired rather than a causal identification method; the learned gate should be interpreted as a relevance proxy and does not represent an identified causal effect.

4-4. Prompt Design

Although spatio-temporal measurements are numerical, a semantic gap remains between continuous signals and the token-based reasoning space of pre-trained language models.

Prior studies, particularly TEST [15] have shown that aligning time-series embeddings with textual prototypes can effectively activate the reasoning capability of frozen large language models. Inspired by this idea, **to address this challenge**, we introduce a textual prefix prompt to provide task-specific semantic priors, helping bridge the gap between numerical spatio-temporal tokens and PLM-based imputation reasoning.

As shown in Figure 4.2, the prompt contains domain information, input structure, missing-value conventions, and task objectives, helping align the PLM with the imputation task.

Given a prompt sequence $\mathbf{p}$, we first tokenize and embed it using the PLM’s tokenizer and embedding layer:

$$\mathbf{E}_{prompt} = \text{Embed}_{PLM}(\text{Tokenizer}_{PLM}(\mathbf{p})) \in \mathbb{R}^{P \times D}, \quad (4.15)$$

where P denotes the prompt length and D is the embedding dimension.

The prompt embeddings are then prepended to the spatio-temporal tokens before being fed into the PLM.

Textual Prefix prompt

[BEGIN PROMPT]

[Domain]: The dataset is a real-world traffic sensor dataset collected from the [dataset] road network.

[Input]: Each row represents a traffic sensor, and each sensor contains a time series of length [seq len].

[Missing Pattern]: Artificially masked values are filled with zero and are accompanied by a binary observation mask.

[Instruction]: Learn temporal dependencies within each sensor and spatial correlations across sensors to infer and fill in the missing traffic values.

[Output]: Return the completed dataset after filling in the missing values.

[END PROMPT]

Figure 4.2: The textual prefix prompt template used in CauST-PLM. The prompt provides dataset semantics, input structure, missing-value conventions, and task-specific instructions to guide PLM-based imputation.

4-5. Bottleneck Encoder-Decoder (BED)

Directly feeding all node tokens into a PLM is inefficient and may amplify noisy observations. To address this, we introduce a bottleneck encoder-decoder module to compress spatial tokens into a smaller set of latent representations.

The gated spatial tokens $\mathbf{S}' \in \mathbb{R}^{N \times D}$ are first compressed into a smaller set of bottleneck

tokens through learnable queries $\mathbf{Q}_b \in \mathbb{R}^{N_b \times D}$, where $N_b < N$:

$$\mathbf{Z} = \text{MHA}(\mathbf{Q}_b, \mathbf{S}', \mathbf{S}'), \quad (4.16)$$

where $\text{MHA}(\cdot)$ denotes the multi-head attention operation. The learnable queries aggregate node-level spatial representations into a compact set of bottleneck tokens, reducing the number of tokens processed by the PLM while preserving global spatial information.

The compressed tokens $\mathbf{Z}$ are then fed into the PLM for global reasoning. After PLM processing, the bottleneck representations are projected back to node-level tokens:

$$\mathbf{H}_n = \text{MHA}(\mathbf{S}', \mathbf{Z}, \mathbf{H}_s), \quad (4.17)$$

where $\mathbf{H}_s$ denotes the PLM outputs corresponding to the bottleneck tokens. Here, the gated spatial tokens $\mathbf{S}'$ serve as the queries, allowing each node representation to attend to the globally refined bottleneck representations.

4-6. Output Projection

After prefix prompting and spatio-temporal token construction, the integrated token sequence is fed into the PLM:

$$[\mathbf{H}_p, \mathbf{H}_t, \mathbf{H}_s] = \text{PLM}([\mathbf{E}_{prompt}; \mathbf{T}_{tok}; \mathbf{Z}]), \quad (4.18)$$

where $\mathbf{H}_p$, $\mathbf{H}_t$, and $\mathbf{H}_s$ denote the hidden representations corresponding to the prompt, temporal, and spatial bottleneck tokens, respectively.

Since the prompt tokens provide task-level context and the temporal tokens summarize global dynamics, the hidden states corresponding to the bottleneck tokens are used for node-level reconstruction through the BED decoder described in Eq. (4.17)

Finally, the imputation output is obtained by adding a residual connection from the original spatial tokens and applying an output MLP:

$$\hat{\mathbf{Y}} = f_{\text{out}}(\text{LN}(\mathbf{H}_n + \mathbf{S}')). \quad (4.19)$$

For imputation, $\hat{\mathbf{Y}} \in \mathbb{R}^{N \times (TC)}$ reconstructs the missing values within the input window, and the training loss is computed only on masked positions.

4-7. Training Objective

To optimize CauST-PLM for imputation, we use an L1 reconstruction loss computed only on the masked positions:

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\Omega|} \sum_{(t,n,c) \in \Omega} |x_{t,n,c} - \hat{x}_{t,n,c}|, \quad (4.20)$$

where Ω denotes the set of masked positions for imputation, $x_{t,n,c}$ is the ground-truth value, and $\hat{x}_{t,n,c}$ is the reconstructed value.

To encourage sparse and informative spatial evidence selection, we further regularize the learned causal gates by penalizing their average activation:

$$\mathcal{L}_{\text{gate}} = \frac{1}{BN} \sum_{b=1}^B \sum_{n=1}^N \beta_{b,n}, \quad (4.21)$$

where B and N denote the batch size and the number of sensors, respectively.

The final training objective is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{gate}}, \quad (4.22)$$

where λ controls the strength of the sparsity regularization.

Chapter5 Experiments

In this section, we present the experimental setup and results to evaluate the performance of our proposed CauST-PLM.

5-1. Datasets

We conduct experiments on four widely used public traffic datasets: **PEMS03**, **PEMS04**, **PEMS07**, and **PEMS08** [3]. These datasets are collected from the Caltrans Performance Measurement System and reflect real-world traffic conditions across California highway networks.

The four datasets exhibit diverse characteristics in terms of network scale, graph density, and traffic dynamics, making them suitable benchmarks for evaluating spatio-temporal imputation models under varying levels of complexity. Their detailed statistics are summarized in Table 5.1.

Table 5.1: Summary of four real-world datasets.

Dataset	#Nodes	#Samples	Sample Rate	Time Range
PEMS03	358	26,208	5 minutes	2018/09–2018/11
PEMS04	307	16,992	5 minutes	2018/01–2018/02
PEMS07	883	28,224	5 minutes	2017/05–2017/08
PEMS08	170	17,856	5 minutes	2016/07–2016/08

5-2. Missing Data Setting

Following previous spatio-temporal imputation studies, we adopt the **Random Point Missing (RPM)** setting for all experiments. Specifically, individual observations are randomly masked while preserving the original temporal and spatial structure of the data.

For each dataset, missing rates of 30%, 50%, 70%, and 90% are considered. A binary mask with the same size as the original spatio-temporal tensor is generated, where observed entries are assigned a value of 1 and randomly selected missing entries are assigned a value of 0. The resulting masked observations are used as the input to the proposed model, while evaluation is conducted only on the artificially masked entries.

5-3. Baseline Methods

To evaluate the effectiveness of CauST-PLM, we compare it with representative baselines for spatio-temporal data imputation, covering statistical, classical time-series, attention-based, and probabilistic spatio-temporal imputation methods. The adopted baselines are summarized as follows.

MEAN. This statistical baseline fills each missing value using the mean of the corresponding historical observations. It does not explicitly model temporal evolution or spatial dependencies, and therefore serves as a simple reference for evaluating whether a model can learn useful structure from the data.

VAR[19]. Vector Autoregression (VAR) is a classical multivariate time-series model that estimates each variable from the past values of all variables. In our setting, it is used to capture linear temporal correlations among sensors and infer the missing entries from the

observed history.

BRITS[1]. Bidirectional Recurrent Imputation for Time Series (BRITS) is a representative recurrent neural network (RNN)-based imputation model. It treats missing values as learnable variables and jointly reconstructs them using bidirectional temporal dependencies through forward and backward recurrent networks. We include BRITS to compare our method against a classical sequential imputation approach that explicitly models temporal dependencies.

mTAN[14]. Multi-Time Attention Network (mTAN) is an attention-based model designed for irregularly sampled time series. It learns continuous-time representations by attending over observed timestamps, making it suitable for handling incomplete temporal observations. We include mTAN to assess the performance of attention-based temporal modeling under missing spatio-temporal data.

CASPER[11]. Causality-Aware Spatio-Temporal Graph Neural Network (CASPER) is a recent spatio-temporal imputation framework that explicitly incorporates causal reasoning into the modeling of spatial and temporal dependencies. It employs a spatio-temporal causal attention (SCA) mechanism to capture informative interactions across neighboring sensors and historical time steps, while suppressing noisy or spurious correlations through causal gating. In addition, CASPER introduces a prompt-based decoder (PBD) to inject global contextual knowledge into the reconstruction process, improving the model’s ability to recover missing values under complex missing patterns.

ST-LLM[12]. Spatio-Temporal Large Language Model (ST-LLM) is a recent framework that adapts large language models (LLMs) for spatio-temporal forecasting tasks. It converts multivariate time-series data into a sequence of tokens and leverages prompt-based learning to align the LLM with downstream prediction objectives. By incorporating temporal context

and structural information into the prompt and token representations, ST-LLM enables the model to capture long-range temporal dependencies and complex correlations.

STD-PLM[8]. Spatial-Temporal Data Modeling with Pre-trained Language Models (STD-PLM) is a recent approach that leverages pre-trained language models (PLMs), such as GPT-style architectures, for spatio-temporal forecasting tasks. It transforms structured traffic data into tokenized representations, including spatial tokens (e.g., node embeddings) and temporal tokens (e.g., time series observations), which are then processed by a PLM to capture complex dependencies.

By utilizing the strong sequence modeling and reasoning capabilities of PLMs, STD-PLM can effectively model long-range temporal dependencies and implicit spatial correlations. However, it does not explicitly incorporate causal-aware mechanisms to filter noisy or spurious spatial interactions, which may limit its robustness under high missing rates. We include STD-PLM as a strong baseline to evaluate the effectiveness of our proposed method.

5-4. Evaluation Metrics

We evaluate imputation accuracy using three standard error metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Following the implementation, the metrics are computed only on the evaluation mask, i.e., the entries that are originally available in the ground truth but hidden from the model during imputation. Let Ω denote this set of evaluated missing positions, where $\Omega = \{(t, n, c) \mid m_{t,n,c}^{ob} = 1, m_{t,n,c}^{cond} = 0\}$. For each evaluated entry, $y_{t,n,c}$ and $\hat{y}_{t,n,c}$ denote the ground-truth value and the imputed value, respectively.

$$\text{MAE} = \frac{1}{|\Omega|} \sum_{(t,n,c) \in \Omega} |y_{t,n,c} - \hat{y}_{t,n,c}|. \quad (5.1)$$

$$\text{RMSE} = \sqrt{\frac{1}{|\Omega|} \sum_{(t,n,c) \in \Omega} (y_{t,n,c} - \hat{y}_{t,n,c})^2}. \quad (5.2)$$

$$\text{MAPE} = \frac{1}{|\Omega|} \sum_{(t,n,c) \in \Omega} \left| \frac{y_{t,n,c} - \hat{y}_{t,n,c}}{y_{t,n,c}} \right|. \quad (5.3)$$

In practice, entries with near-zero ground-truth values are excluded from MAPE computation to avoid numerical instability, consistent with the thresholding behavior in the metric implementation. Lower values of all three metrics indicate better imputation performance.

5-5. Implementation Details

For each dataset, we split the data into training, validation, and test sets with a ratio of 6:2:2. To evaluate the robustness of CauST-PLM under different sparsity levels, we construct Random Point Missing (RPM) settings with missing rates of 30%, 50%, 70%, and 90%.

The observed entries serve as conditional inputs, while the masked entries are treated as evaluation targets. All masked values are replaced with zero and accompanied by a binary mask tensor, enabling the model to distinguish observed and missing positions. For fair comparison, all models are evaluated under the same missing patterns for each dataset and missing rate.

The output length is set equal to the input window length, so the model reconstructs missing values within the observed window rather than forecasting future timestamps. The loss and evaluation metrics are computed only on masked positions to ensure that performance reflects the imputation capability.

We implement our model in PyTorch and optimize it using AdamW with an initial learning rate of 0.001 and a batch size of 64 for all experiments. We adopt GPT-2 as the backbone pre-trained language model and utilize its first three Transformer layers throughout all experiments. The hidden dimension of the PLM is 768. The constructed prompt, temporal, and spatial representations are directly provided to GPT-2 through the input embedding interface. To efficiently adapt the PLM to the spatio-temporal imputation task, all GPT-2 parameters are initially frozen. LoRA is then applied to the attention projection modules with a rank of 16, a scaling factor of 32, and a LoRA dropout rate of 0. In addition, the LayerNorm layers and positional embeddings are updated during training, while all remaining GPT-2 parameters remain frozen. All experiments are conducted on a Linux workstation equipped with an Intel i7-13700K CPU, an NVIDIA RTX 4090 GPU, and 64 GB RAM.

5-6. Main Result

Table 5.2 presents the imputation results under different Random Point Missing (RPM) rates across four traffic datasets. Overall, CauST-PLM achieves the best overall trade-off among all compared methods, demonstrating strong robustness under varying missing ratios.

Against traditional statistical baselines, including MEAN and VAR, CauST-PLM consistently achieves substantially lower errors across all datasets. MEAN yields the largest errors since it ignores both temporal variation and spatial dependencies, while VAR improves by modeling multivariate temporal correlations but remains limited by its linear assumptions. These results indicate that nonlinear spatio-temporal modeling is essential for realistic traffic imputation.

Compared with the recurrent baseline BRITS, CauST-PLM consistently achieves lower

errors across all datasets and missing rates. BRITS reconstructs missing values by exploiting bidirectional temporal dependencies, but it does not explicitly model the graph-structured spatial relationships among traffic sensors. Its performance also degrades noticeably as the missing rate increases, suggesting that relying mainly on temporal recurrence becomes insufficient when only limited observations are available. In contrast, CauST-PLM jointly incorporates temporal context, spatial representations, and PLM-based global reasoning, leading to more robust reconstruction under different missing conditions.

Relative to the attention-based baseline mTAN, CauST-PLM generally achieves better performance in most settings. Although mTAN models temporal irregularity effectively, it lacks explicit graph-aware spatial modeling, which limits its ability to capture sensor interactions in traffic networks.

CASPER, which introduces lightweight causal gating without PLM-based structured reasoning, CauST-PLM consistently achieves stronger performance under most settings. This suggests that causal-aware spatial refinement alone is beneficial, while integrating it into a more expressive PLM framework further improves reconstruction quality.

Among the LLM-based baselines, including ST-LLM and STD-PLM, CauST-PLM achieves comparable and often slightly better performance across most datasets and missing rates. Since all three methods leverage PLM-based representations, the performance gaps are relatively smaller than those of traditional baselines. Nevertheless, the improvements indicate that the proposed causality-aware spatial filtering can further refine spatial tokens by suppressing noisy or weakly relevant sensor interactions before PLM reasoning.

The advantage of CauST-PLM becomes more evident under higher missing rates. When the missing rate increases to 70% and 90%, most baselines show larger performance degradation due to limited observations. In contrast, CauST-PLM maintains relatively stable

performance, indicating that the combination of hard prompt guidance, structured spatio-temporal tokenization, and causal-aware spatial refinement improves robustness under sparse observation settings.



Table 5.2: Performance comparison results of all models on all datasets. The best results are highlighted in bold, and the second-best results are underlined.

Datasets	Methods	Missing rate 30%			Missing rate 50%			Missing rate 70%			Missing rate 90%		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
PEMS03	MEAN	40.164	51.216	110.13%	40.169	51.303	110.19%	40.231	51.591	110.34%	40.648	51.703	110.52%
	VAR	15.909	25.650	16.97%	16.959	27.520	17.92%	18.286	29.462	19.42%	23.058	36.718	25.24%
	BRITS	13.975	26.947	16.97%	15.350	28.417	18.86%	17.593	31.015	22.36%	23.504	38.681	31.67%
	mTAN	14.440	24.051	19.84%	15.577	26.082	20.99%	17.180	28.998	21.89%	18.799	30.836	24.54%
	CASPER	15.648	26.805	16.67%	16.074	28.536	15.68%	17.917	32.185	20.87%	31.775	56.367	49.04%
	ST-LLM	<u>12.641</u>	21.134	15.47%	14.254	24.103	18.06%	16.487	27.764	20.29%	22.729	38.808	23.89%
	STD-PLM	12.664	22.351	<u>13.36%</u>	<u>13.640</u>	22.653	<u>14.43%</u>	<u>14.688</u>	24.283	<u>15.35%</u>	<u>18.805</u>	<u>31.489</u>	<u>19.62%</u>
	CauST-PLM	12.490	<u>21.917</u>	13.06%	13.256	<u>22.770</u>	13.79%	14.615	<u>25.702</u>	15.10%	18.112	31.141	18.23%
PEMS04	MEAN	46.374	59.853	71.11%	46.454	59.911	71.19%	46.617	60.196	71.27%	46.699	60.273	71.35%
	VAR	19.736	32.211	13.97%	20.424	33.336	14.53%	21.438	34.827	15.46%	25.011	39.188	19.57%
	BRITS	21.574	36.573	16.04%	22.682	38.062	17.27%	24.289	39.976	19.18%	31.012	47.894	28.03%
	mTAN	20.751	33.687	21.27%	22.038	35.482	22.31%	23.268	36.678	26.85%	25.932	40.990	28.21%
	CASPER	19.293	32.117	13.36%	21.195	33.606	16.75%	21.618	34.979	15.82%	24.857	39.083	19.34%
	ST-LLM	17.374	27.831	12.43%	18.475	30.048	12.98%	19.616	32.726	14.86%	22.427	36.439	15.43%
	STD-PLM	<u>16.868</u>	<u>27.483</u>	<u>11.28%</u>	<u>17.529</u>	<u>28.427</u>	<u>11.68%</u>	<u>18.523</u>	<u>30.357</u>	<u>12.33%</u>	<u>21.492</u>	35.272	<u>14.31%</u>
	CauST-PLM	16.749	27.334	11.15%	17.321	28.191	11.57%	18.419	30.110	12.23%	21.445	<u>35.388</u>	14.13%
PEMS07	MEAN	48.376	66.337	63.91%	48.39	66.403	63.98%	48.415	66.488	64.05%	48.447	66.506	64.11%
	VAR	36.905	57.855	15.65%	39.120	61.971	16.69%	39.559	62.482	17.38%	42.309	66.232	20.04%
	BRITS	27.480	51.258	13.25%	29.450	53.358	14.66%	33.146	58.026	16.92%	40.025	64.787	22.16%
	mTAN	21.372	33.065	13.79%	22.632	35.066	12.92%	24.919	38.662	13.84%	28.304	43.955	14.72%
	CASPER	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
	ST-LLM	<u>16.901</u>	<u>27.082</u>	7.89%	18.955	31.125	9.09%	21.611	36.874	11.15%	27.077	48.646	12.40%
	STD-PLM	16.958	27.333	<u>7.24%</u>	<u>17.628</u>	<u>28.428</u>	7.56%	19.109	31.176	8.13%	24.146	<u>43.452</u>	10.30%
	CauST-PLM	16.753	25.956	7.16%	17.603	28.284	<u>7.59%</u>	<u>19.120</u>	<u>31.302</u>	<u>8.17%</u>	<u>24.346</u>	43.228	<u>10.50%</u>
PEMS08	MEAN	30.561	64.256	60.61%	30.573	64.283	60.62%	30.579	64.302	60.67%	30.588	64.312	60.67%
	VAR	18.367	28.768	13.00%	20.467	32.738	14.54%	22.747	35.845	16.42%	26.937	41.393	20.36%
	BRITS	17.229	28.098	11.58%	18.850	30.934	12.74%	21.253	34.832	14.31%	27.837	44.985	19.43%
	mTAN	15.662	24.153	12.88%	16.467	25.232	13.25%	18.180	27.905	14.55%	21.144	32.096	19.78%
	CASPER	13.647	19.841	9.51%	14.621	23.016	10.98%	15.215	25.628	11.47%	19.451	34.908	12.72%
	ST-LLM	13.686	21.205	9.88%	14.794	23.008	11.11%	16.599	26.256	12.23%	19.997	33.890	12.91%
	STD-PLM	12.362	19.908	8.10%	13.143	<u>21.294</u>	<u>8.96%</u>	<u>14.557</u>	<u>23.274</u>	<u>9.72%</u>	18.619	31.467	12.22%
	CauST-PLM	<u>12.618</u>	<u>19.942</u>	<u>8.23%</u>	<u>13.181</u>	20.832	8.80%	14.478	23.271	9.68%	<u>18.696</u>	<u>31.468</u>	<u>12.33%</u>

5-7. Ablation Study

To validate the effectiveness of each component in our proposed model, we conduct ablation studies by removing or modifying key modules of CauST-PLM and comparing their imputation performance under the same experimental settings.

Specifically, we evaluate four variants. **w/o PLM** replaces the pre-trained language model with a Transformer encoder of the same number of layers and hidden dimension, allowing us to evaluate whether the performance gain comes from pre-trained knowledge rather than model capacity. **w/o SCA** removes the spatial-temporal causal attention module, allowing the model to use the original spatial tokens without causal gating. **w/o prompt** removes the textual prefix prompt, so the PLM receives only the continuous temporal and spatial embeddings as input. **w/o BED** removes the bottleneck encoder-decoder module and directly feeds all spatial tokens into the PLM without latent compression.

The results are summarized in Table 5.3 and Table 5.4. The full model achieves the best overall trade-off across different datasets and missing rates, although individual metrics may fluctuate slightly.

The comparison between the full model and **w/o PLM** demonstrates that the performance gain is not solely due to the Transformer architecture, but also stems from the transferable knowledge acquired during large-scale pre-training. Replacing the PLM with a Transformer encoder generally degrades reconstruction performance, and the performance gap becomes substantially larger under extremely high missing rates. This observation suggests that pre-trained knowledge becomes increasingly important when the available observations are limited, enabling the model to better exploit temporal patterns for missing-value reconstruction.

Among the four variants, **w/o SCA** generally causes a larger performance degradation than **w/o prompt**, while **w/o BED** mainly affects the computational feasibility rather than the reconstruction accuracy. This indicates that causality-aware spatial filtering is important for selecting informative sensor tokens and suppressing noisy or weakly related spatial correlations before PLM reasoning. Without SCA, the model directly relies on the original spatial tokens, which may increase the influence of irrelevant sensor interactions and reduce reconstruction robustness. In contrast, removing the bottleneck encoder-decoder has a relatively small impact on reconstruction accuracy for the smaller PEMS08 dataset, but leads to GPU out-of-memory (OOM) on the larger PEMS03 dataset. This suggests that the bottleneck encoder-decoder primarily improves the computational efficiency and scalability of the proposed framework, enabling PLM-based imputation on larger sensor networks.

The **w/o prompt** variant also performs worse than the full model in most cases, showing that textual task guidance helps the PLM understand the imputation objective and the missing-value convention. Compared with SCA, however, the effect of the prompt is more moderate, suggesting that the prompt mainly improves task alignment while SCA contributes more directly to the quality of spatial representation.

The **w/o BED** variant highlights the importance of the bottleneck encoder-decoder from a computational perspective. Although removing BED remains feasible on the smaller PEMS08 dataset and even yields slightly better performance on some metrics, the model fails to train on the larger PEMS03 dataset due to GPU memory exhaustion (OOM). This result indicates that the bottleneck encoder-decoder is not primarily introduced to improve accuracy, but rather to reduce the computational and memory cost of PLM processing, making the proposed framework scalable to larger sensor networks. Overall, these observations indicate that all four components play complementary roles in the proposed framework. The pre-

trained language model provides transferable knowledge for robust spatio-temporal reasoning, the SCA module refines spatial representations by suppressing noisy sensor interactions, the textual prompt improves task alignment, and the bottleneck encoder-decoder reduces computational and memory costs, making the proposed framework scalable to larger sensor networks.

Table 5.3: Ablation study on PEMS03 under different missing rates.

Method	30%			50%			70%			90%		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
w/o PLM	13.078	20.598	8.65%	13.768	21.628	9.14%	15.428	25.137	10.90%	25.734	46.125	23.50%
w/o SCA	12.561	22.375	13.16%	13.332	22.823	13.78%	14.867	25.261	15.05%	19.784	34.001	20.51%
w/o prompt	12.556	22.069	13.10%	13.278	22.535	13.87%	14.755	25.654	14.93%	18.541	32.107	18.96%
w/o BED	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
CauST-PLM	12.490	21.917	13.06%	13.256	22.77	13.79%	14.615	25.702	15.10%	18.112	31.141	18.23%

Table 5.4: Ablation study on PEMS08 under different missing rates.

Method	30%			50%			70%			90%		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
w/o PLM	12.802	22.941	13.64%	13.879	23.113	14.08%	15.575	27.023	17.91%	36.462	64.280	60.71%
w/o SCA	12.697	20.292	8.54%	13.247	21.023	8.81%	14.868	23.434	9.88%	19.437	32.243	13.36%
w/o prompt	12.678	20.332	8.45%	13.298	21.063	8.84%	14.796	22.373	9.74%	18.949	31.656	12.53%
w/o BED	12.655	20.180	8.15%	13.192	20.193	8.58%	14.349	22.954	9.40%	18.322	31.070	11.84%
CauST-PLM	12.618	19.942	8.23%	13.181	20.832	8.80%	14.478	23.271	9.68%	18.696	31.468	12.33%

5-8. Parameter Study

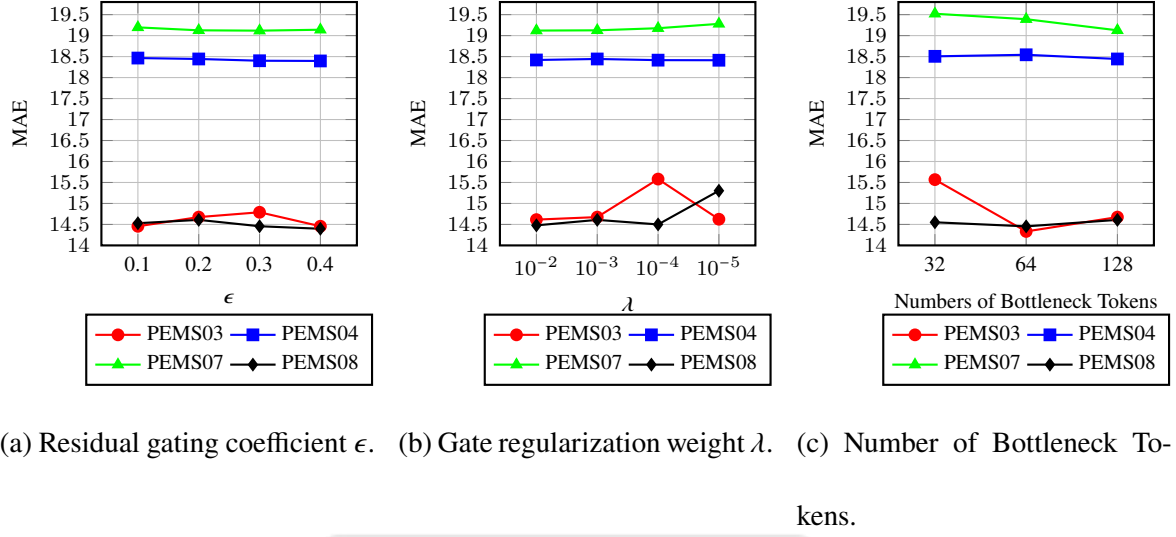


Figure 5.1: Hyper-parameter analysis of CauST-PLM. (a) Residual gating coefficient ϵ . (b) Gate regularization weight λ . (c) Number of Bottleneck Tokens.

We further investigate the influence of three key hyper-parameters, including the residual gating coefficient ϵ , the gate regularization weight λ , and the Number of Bottleneck Tokens. These parameters respectively control the degree of spatial information preservation after causal-inspired filtering, the sparsity of spatial token selection, and the compression capacity of the bottleneck representations before PLM reasoning. The results are illustrated in Fig. 5.1(a), Fig. 5.1(b), and Fig. 5.1(c). Overall, the performance variations under different settings are relatively small, indicating that the proposed model is robust and does not rely on highly sensitive hyper-parameter tuning.

For the residual gating coefficient, ϵ controls the balance between causal-inspired spatial filtering and the preservation of original node-specific information. In traffic sensor networks, although some sensor interactions may be noisy or weakly related, individual sensors can still

contain useful local traffic patterns. Therefore, excessively small ϵ values may over-suppress potentially informative spatial representations, whereas overly large ϵ values weaken the filtering effect by retaining too much original information. Although different datasets exhibit slightly different preferences, the overall performance remains stable across the tested range. We therefore adopt $\epsilon = 0.2$, which provides a balanced trade-off between suppressing noisy spatial interactions and preserving useful spatial information across all datasets.

As for the gate regularization weight, λ controls the sparsity of the learned spatial gates. Traffic sensor networks often contain many correlated sensors, yet not all spatial interactions are equally informative for reconstructing a particular missing value. Without sufficient regularization, the model may retain excessive or redundant spatial information, reducing the effectiveness of the filtering module. Conversely, overly strong regularization may suppress useful neighboring sensors. Among the evaluated settings, $\lambda = 0.01$ achieves the best overall performance on most datasets and remains highly competitive on the others. This suggests that an appropriate sparsity constraint encourages the model to focus on a compact set of informative spatial representations while suppressing noisy or redundant sensor interactions.

Finally, the number of bottleneck tokens controls the compression capacity of the spatial representations before PLM reasoning. This design is particularly important for traffic sensor networks, where the number of sensors varies substantially across datasets and directly affects the computational cost of attention-based sequence modeling. Using too few bottleneck tokens may over-compress node-level information and limit the representation of diverse spatial traffic patterns, whereas using too many tokens increases computational and memory costs while providing only limited additional information. As shown in Fig. 5.1(c), increasing the number of bottleneck tokens from 32 to 64 generally improves the imputation accuracy, whereas further increasing it to 128 yields only marginal gains on some datasets while

introducing additional computational cost. Therefore, we adopt a Number of Bottleneck Tokens of 64 as a good trade-off between representation capacity and computational efficiency, providing sufficient spatial representation while maintaining scalability for larger traffic sensor networks.

5-9. Case Study

Table 5.5: Top-10 High- β and Low- β Sensors.

Top-10 High- β Sensors		Top-10 Low- β Sensors	
Sensor Index	β (Gate Value)	Sensor Index	β (Gate Value)
243	0.92	12	0.03
87	0.91	289	0.04
198	0.89	301	0.05
52	0.87	305	0.06
310	0.85	15	0.06
121	0.83	288	0.07
76	0.82	308	0.07
199	0.80	9	0.08
33	0.79	311	0.08
145	0.78	6	0.09

To provide a qualitative analysis of the proposed SCA module, we analyze the learned gate values on a test sample from the PEMS03 dataset under 70% missing setting. Table 5.5 lists the top-10 sensors with the highest and lowest gate values. As can be observed, the learned gate values vary considerably across different sensors. The highest gate values range from 0.78 to 0.92, whereas the lowest gate values are between 0.03 and 0.09, indicating that

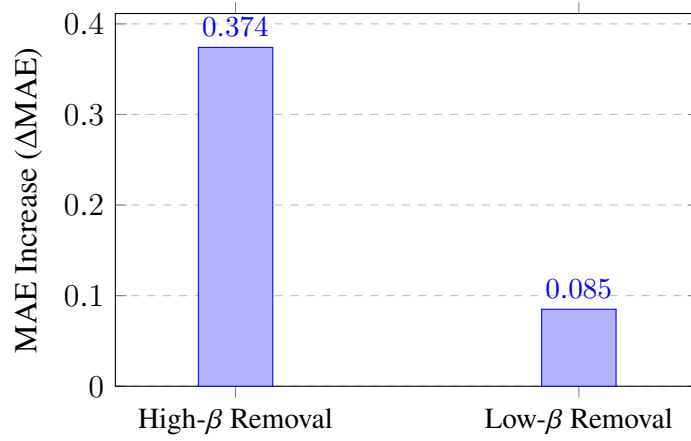


Figure 5.2: MAE increase after removing high- β and low- β sensors on PEMS03 under the 70% missing setting.

the proposed SCA module selectively preserves spatial information rather than treating all sensors equally.

Sensors assigned higher gate values retain a larger proportion of their spatial representations during the residual gating process and therefore contribute more to the subsequent PLM reasoning. In contrast, sensors with lower gate values are more strongly suppressed, suggesting that their spatial information is considered less informative under the current traffic condition.

To further investigate whether the learned gate values reflect the actual contribution of spatial information to imputation, we conduct a sensor removal analysis. Specifically, the spatial representations of the top-10 sensors with the highest and lowest β values are separately removed during inference, and the resulting increase in MAE is measured relative to the original model.

As shown in Figure 5.2, removing the high- β sensors increases the MAE by 0.374, whereas removing the low- β sensors results in a substantially smaller increase of 0.085. The degradation caused by removing high- β sensors is approximately 4.4 times larger than that

caused by removing low- β sensors. This result indicates that spatial representations assigned higher gate values have a greater influence on the reconstruction process, while those assigned lower gate values contribute relatively less.

Together with the gate value distribution shown in Table 5.5, the removal analysis provides further evidence that the proposed SCA module performs adaptive spatial filtering rather than merely assigning arbitrary gate values. In particular, SCA tends to preserve spatial representations that are more influential for imputation while suppressing less relevant spatial information before PLM reasoning. This observation is consistent with the design objective of the proposed causal-inspired spatial filtering mechanism.



Chapter6 Conclusion

In this paper, we proposed CauST-PLM, a causality-aware spatio-temporal pre-trained language model for imputation. The model converts incomplete spatio-temporal observations into temporal and spatial tokens, injects missing-value information through mask-aware tokenization, and uses textual prefix prompting to align the PLM with the imputation objective. To improve the quality of spatial evidence, CauST-PLM further introduces a spatial-temporal causal attention module that filters noisy or weakly relevant sensor information before PLM reasoning.

Extensive experiments on four public **PEMS traffic flow datasets** (PEMS03, PEMS04, PEMS07, and PEMS08) demonstrate that CauST-PLM achieves strong imputation performance under different Random Missing rates. Compared with statistical, classical time-series, attention-based, and representative PLM-based baselines, CauST-PLM obtains lower errors in most settings, showing its ability to model nonlinear spatio-temporal dependencies. In particular, compared with STD-PLM, a representative PLM-based spatio-temporal imputation method, the proposed model provides comparable or better performance in most cases, indicating that causality-aware spatial refinement can further enhance PLM-based imputation. The ablation studies also demonstrate the complementary roles of the proposed components. Among them, SCA plays a particularly important role in improving spatial representation quality, while the prompt guidance, bottleneck encoder-decoder, and PLM further enhance task alignment, computational scalability, and reconstruction performance. The comparison with the Transformer-based variant further indicates that leveraging pre-trained knowledge contributes to more robust spatio-temporal imputation, especially under high missing rates.

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